



Jirawat Kritmasup Mechanical Engineering

Contact

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Skills

Proficiency in C/C++, Python, and embedded firmware development. Experience with OpenCV, object detection, and stereo vision. Strong understanding of microcontroller-based control systems. Knowledge of electronics, sensors, and actuators. Familiarity with real-time systems and signal processing. Basic experience in machine learning or AI-based vision is a plus

Work History

Research and Development Engineer

Kasetsart University Kamphaeng Saen Campus
Nakhon Pathom

Responsible for developing and integrating robotic systems using microcontrollers and computer vision technologies. Focused on creating intelligent robots capable of object detection, depth perception, and autonomous decision-making through vision-based algorithms.

- Design, develop, and program robotic systems using microcontrollers (e.g., ESP32, STM32, Arduino, etc.).
- Implement and optimize object detection and tracking using OpenCV and deep learning frameworks.
- Integrate and calibrate stereo camera systems for depth estimation and 3D perception.
- Develop control algorithms for motor control, navigation, and sensor fusion.
- Interface hardware and software modules including motors, encoders, cameras, and communication interfaces (UART, SPI, I2C, etc.).
- Conduct testing, debugging, and performance tuning of robotic prototypes.
- Collaborate with mechanical and software teams to achieve full robot functionality.
- Document development processes, experiments, and technical findings.

Education

2018-5

Bachelor of Engineering: Mechanical Engineering

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Nakhon Pathom